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Methods for generating mimetic innate immune cells from pluripotent stem cells

Abstract

Human pluripotent stem cells (hPSCs), especially induced pluripotent stem cells (iPSCs) provide a promising starting material to produce mimetic innate immune cells such as natural killer (NK) cells and $\gamma\delta$ T-cells for cancer immunotherapy. To facilitate consistent mass production, an overall manufacturing scheme to make mimetic innate immune cells from hPSCs was designed and demonstrated. Particularly, a robust protocol to differentiate hPSCs into NK cells or $\gamma\delta$ T-cells through sequential hematopoietic differentiation on stromal cell line deficient in expressing M-CSF and lymphoid commitment on stromal cell line deficient in expressing M-CSF ectopically expressing DLL1 without employing CD34⁺ cell enrichment and spin embryoid body formation is established. Using this two-stage protocol, the generation of functional mimetic NK cells and functional mimetic $\gamma\delta$ NKT-cells was demonstrated from hPSCs, including hESCs, peripheral blood cell-derived iPSCs (PBC-iPSCs). non-T cell-derived iPSCs or $\gamma\delta$ T cell-derived iPSCs and the use of these mimetic innate immune cells in killing cancer cells.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage filing under 35 U.S.C. § 371 of International Patent Application Serial No. PCT/SG2018/050051, filed Feb. 7, 2018, which claims the priority to Singapore Application No. 10201700937P, filed Feb. 7, 2017 and Singapore Application No. 10201705582S, filed Jul. 6, 2017, the contents of each of which are incorporated herein by reference.

FIELD

(2) The present invention relates to methods and kits for generating innate immune cells, preferably natural killer cells or gamma delta T-cells and methods of using the innate immune cells in treatment and methods of generating induced pluripotent stem cells.

BACKGROUND

(3) In 2015, about 90.5 million people had cancer and about 14.1 million new cases occur each year (GBD 2015, Lancet. 388 (10053) 1545-1602). Cancer is often treated with some combination of radiation therapy, surgery, chemotherapy, and/or targeted therapy. While there are many treatments available 8.8 million people die from cancer each year and there is a constant need for new treatments.

(4) Cancer immunotherapies exploiting innate immune cells are the new hope for cancer treatment (Woo et al. *Annual review of immunology* 33, 445-474 (2015)). These innate immune cells including natural killer (NK) cells and $\gamma\delta$ T cells can recognize a wide range of cancer cells through major histocompatibility complex (MHC)-independent mechanisms (Scheper, et al. *Leukemia* 28, 1181-1190 (2014)). This unique feature allows the use of innate immune cells to treat cancers in many recipients without MHC-restriction of conventional $\alpha\beta$ T cell-based strategies. Currently, healthy donor-derived blood cells are the commonly used cell source to generate a large number of innate immune cells (Kondo, et al. *Cytotherapy* 10, 842-856 (2008)). However, these donor blood cell-dependent platforms are challenging for centralized, standardized and large-scale production due to the use of variable and limited starting materials and the complicated logistics.

(5) Gamma delta T-cells ($\gamma\delta$ T cells) are innate immune cells that recognize cancer cells via major histocompatibility complex-independent mechanisms. This feature allows the use of donor-derived $\gamma\delta$ T cells to treat cancers in different patients. $\gamma\delta$ T cells account for 1-10% of peripheral blood lymphocytes. Among these circulating $\gamma\delta$ T cells, V γ 9V δ 2 T cells are the major subset that reacts to phosphoantigens through their V γ 9V δ 2 T-cell receptors (TCRs), and thus recognizes infected cells or malignant cells. During in vivo development, somatic recombination of TCRG and TCRD genes is an early crucial step for $\gamma\delta$ cells to obtain their TCR γ chain and δ chain. To generate $\gamma\delta$ T cells from iPSCs, a logical strategy would be to accurately recapitulate this process of somatic recombination of TCRG and TCRD genes during in vitro differentiation to produce functional $\gamma\delta$ TCRs. But such a strategy remains very difficult for current technology.

(6) Natural killer cells (NK cells) are lymphocytes of the innate immune system that are able to kill a broad spectrum of malignant cells and virus-infected cells (Domogala, et al. *Frontiers in immunology* 6, 264 (2015)). The target recognition and activation of NK cells depend on an array of activating receptors and inhibitory receptors, which are different from the MHC-restricted $\alpha\beta$ T-cell receptor (TCR)-dependent mechanism of $\alpha\beta$ T cells (Moretta, et al. *Frontiers in immunology* 5, 87 (2014)). Thus, it is possible to use allogeneic NK cells to treat cancer with less chance of causing graft-versus-host-disease (GVHD) (Leung. *Clinical cancer research: an official journal of the American Association for Cancer Research* 20, 3390-3400 (2014)). The distinct target recognition mechanism of NK cells significantly extends the potential cell sources that can be used in adoptive immunotherapy for cancer beyond autologous NK cells.

(7) In current clinical trials, large dosages of NK cells ranging from 5×10^8 - 5×10^7 /kg body weight are required (Lapteva et al. *Critical reviews in oncogenesis* 19, 121-132 (2014)). One typical approach to derive such large amounts of allogeneic NK cells is NK cell enrichment from

large volumes of donor peripheral blood. Starting from donor-derived leukapheresis products, different protocols have been established to produce NK cell products by immunomagnetic depletion of T cells and B cells and selection of CD56⁺ cells (Koehl, et al. *Frontiers in oncology* 3, 118 (2013)). The purities of such products are especially crucial for clinical application in an allogeneic setting, in which the residual T cells may cause GVHD and prohibit the infusion of high dosages. Current generation of high-purity NK cell products requires a prolonged procedure that not only compromises the recovery of NK cells but also affects their viability and potency. With a low recovery rate, it is difficult to obtain sufficient NK cells from a single leukapheresis product. Another popular approach to produce NK cell therapeutics is NK cell expansion using feeder cells, such as K562 cells modified with membrane-bound molecules such as interleukin (IL)-15 and 4-1BB ligand (K562-mb15-41BBL) (Fujisaki, et al. *Cancer research* 69, 4010-4017 (2009)). These feeder cells can rapidly expand the NK cells from peripheral blood mononuclear cells (PBMCs) by 21.6-fold (Fujisaki, et al.) or from cryopreserved apheresis products by 70-fold (Lapteva et al.). Yet the purities of such NK cell products are about 70% and further, time consuming methods of depletion of concomitant CD3⁺ T cells is necessary for allogeneic use (Lapteva et al.). The above-mentioned two approaches are feasible to generate NK cell therapeutics from donor cells, but like any other donor cell-dependent manufacturing processes, generating NK cell products from primary cells of various donors is difficult to be standardized due to the variable starting materials. It demands particular skills and facilities, complicated logistics and high operation cost, thus it is not amenable to bulk manufacturing.

(8) Currently, healthy donor-derived blood cells are the commonly used cell source to generate a large number of innate immune cells. However, these donor blood cell-dependent platforms are challenging for centralized, standardized and large-scale production due to the use of variable and limited starting materials and the complicated logistics. Another hurdle faced by current strategies for NK cell production is that NK cells express clonally distributed inhibitory receptors known as killer cell immunoglobulin-like receptors (KIRs) (Parham, 2005 *Nat. Rev. Immunol.* 5, 201-214). Each individual KIR recognizes a specific human leukocyte antigen (HLA) class I molecule known as KIR ligand, e.g., KIR2DL1 binds HLA-C2, KIR2DL2 and KIR2DL3 bind HLA-C1, KIR3DL1 binds HLABw4, and KIR3DL2 binds HLA-A3 and HLA-A11 (Thielens et al., 2012 *Curr. Opin. Immunol.* 24, 239-245). Binding of KIR ligands to inhibitory KIRs suppresses cytotoxicity of NK cells. To alleviate such inhibition on NK cells and thus to enhance their cytotoxicity against a patient's cancer cells, elaborately selecting an NK cell donor for that particular patient to obtain a KIR-HLA mismatch in an anti-cancer direction is critical (Thielens et al., 2012). This selection is based on donor KIR typing and recipient HLA typing. A donor is suitable if an inhibitory KIR is present in the donor but the KIR ligand is absent in the recipient (Leung, 2014 *Clin. Cancer Res.* 20, 3390-3400). The involvement of both KIR and HLA, two highly diverse gene families in human immune system, decides that current donor cell-dependent manufacturing platforms can only produce “custom-made” NK cell products for limited patients instead of “off-the-shelf” ones for a wide range of patients. Therefore, it is imperative to explore alternative manufacturing strategy that circumvents these aforementioned issues.

(9) In the age of pluripotency, human pluripotent stem cells (hPSCs), especially induced pluripotent stem cells (iPSCs) have emerged as a reliable and standardizable starting material to generate immune cells like dendritic cells (Zeng, et al. *Journal of immunology* 188, 4297-4304 (2012)) as well as NK cells (Woll, et al. *Blood* 113, 6094-6101 (2009)). Although a couple of protocols have been developed to generate NK cells from hPSCs, the requirements of enriching CD34⁺ cells (Woll, et al. *Blood* 113, 6094-6101 (2009)) or spin embryoid body (EB) formation (Knorr, D. A. et al. *Stem cells translational medicine* 2, 274-283 (2013)) determine that these existing protocols are very complex and time consuming thus more suitable for laboratory use rather than industrial manufacturing. Moreover, using these current methods, the resulting NK cells express high-level KIRs (Knorr et al., 2013), which restrict their applications to recipients of certain HLA typing. Up

to now a robust protocol to consistently generate NK cells from various hPSC sources is yet to be established.

(10) Inducing hematopoietic differentiation of hPSCs is a crucial first step for generating NK cells from hPSCs. One approach is to co-culture hPSCs with bone marrow stromal cells such as S17 and M2-10B4 to obtain CD34⁺ hematopoietic precursors (Woll, et al. Blood 113, 6094-6101 (2009)). However, this method requires a prolonged co-culture of up to 21 days and a subsequent sorting of CD34⁺ cells before their further differentiation into NK cells. This sorting step will not only complicate the protocol, but also affect the yield of NK cells due to the exclusion of many other hematopoietic progenitors that also have NK cell differentiation potential. Another approach to generate hematopoietic cells is to form EBs (Tabatabaei-Zavareh, et al. PloS one 2, e232 (2007)) and recently spin EBs Knorr, (D. A. et al. Stem cells translational medicine 2, 274-283 (2013)). This random differentiation approach, however, is highly hPSC line-dependent and thus not robust enough for various hPSC sources. Moreover, the spin EBs formation requires a prior time-consuming adaptation of hPSCs to TrypLE digestion for at least 10 passages, which may not work for every hPSC line. These TrypLE-adapted hPSCs are then seeded into 96-well plate at a density of 3,000 cells per well and spun down for spin EB formation. This is followed by harvesting and seeding the spin EBs at a density of 6 spin EBs per well in 24-well plate for further NK cell differentiation. Apparently, this labor-intensive and skill-demanding spin EB approach is difficult for scale-up in commercial production.

(11) Co-culturing with stromal cells to promote further differentiation of hematopoietic progenitors is the second step for generating NK cells from hPSCs. Several stromal cell lines have been used for coculturing. Although stromal cell lines such as MS-5 and AFT024 have been proved to support NK cell development, the resulting cells are a heterogeneous population containing both CD56⁺ and CD56⁻ cells (Woll, et al. Blood 113, 6094-6101 (2009)). A stromal cell line EL08-1D2 has showed better support for NK cell development (McCullar, V. et al. Experimental hematology 36, 598-608 (2008)); however, its efficacy has only been demonstrated with sorted CD34⁺ cells or spin EBs (Woll, et al. Blood 113, 6094-6101 (2009)).

SUMMARY

(12) An object of the invention is to ameliorate some of the above mentioned difficulties preferably by using a reliable, unlimited and standardizable starting cell source of human pluripotent stem cells such as induced pluripotent stem cells (iPSCs) to generate mimetic innate immune cells for use in immunotherapy treatment.

(13) Accordingly, a first aspect of the invention includes a method of generating mimetic innate immune cells, the method comprising: (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor to generate progenitors of innate immune cells; (b) co-culturing the progenitors of innate immune cells with a stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(14) Another aspect of the invention includes mimetic innate immune cells comprising cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL NKG2A/CD94 or a combination thereof and with low or no expression of killer cell immunoglobulin-like receptors (KIR).

(15) Another aspect of the invention includes mimetic innate immune cells as described herein for use in a treatment.

(16) Another aspect of the invention includes mimetic innate immune cells as described herein for use in treating cancer.

(17) Another aspect of the invention includes use of mimetic innate immune cells as described

herein for manufacture of a medicament in the treatment of cancer.

(18) Another aspect of the invention includes a kit for generating mimetic innate immune cells, the kit comprising: (a) a human pluripotent stem cell line; (b) a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF); (c) a stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta like 1 (DLL1); (d) a first simple media for co-culturing a human pluripotent stem cells with the stromal cell line deficient in expressing M-CSF; and (e) a second media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15).

(19) Another aspect of the invention includes a method of treating a patient with cancer comprising: administering to the patient mimetic innate immune cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL NKG2A/CD94 or a combination thereof and with low or no expression of killer cell immunoglobulin-like receptors KIR generated by, (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor to generate progenitors of innate immune cells; (b) co-culturing the progenitors of innate immune cells with a stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(20) Another aspect of the invention includes a method of generating induced pluripotent stem cells (iPSC) from non-T cells of peripheral blood cells, comprising: (a) incorporating reprogramming transcription factors into non-T cells; (b) growing the non-T cells incorporated with the reprogramming transcription factors to become pluripotent stem cells.

(21) Another aspect of the invention includes a method of generating induced pluripotent stem cells (iPSC) from gammadelta T cells ($\gamma\delta$ T cells), comprising: (a) incorporating reprogramming transcription factors into $\gamma\delta$ T cells; (b) growing the $\gamma\delta$ T cells incorporated with the reprogramming transcription factors to become pluripotent stem cells.

(22) Another aspect of the invention includes an induced pluripotent stem cells (iPSC) comprising a TCR gene arranged in a TCRG gene configuration or a TCR gene arranged in a TCRD gene configuration.

(23) Another aspect of the invention includes a method of generating mimetic NK cells, the method comprising: (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) to generate primed innate immune cells; (b) co-culturing the primed innate immune cells with a stromal cell line deficient in expressing M-CSF and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(24) Another aspect of the invention includes a method of generating mimetic $\gamma\delta$ T cells, the method comprising: (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) to generate primed innate immune cells; (b) co-culturing the primed innate immune cells with a stromal cell line deficient in expressing M-CSF and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(25) Another aspect of the invention includes mimetic NK cells comprising cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 or a combination thereof, with no CD3 expression and with low or no expression of killer cell

immunoglobulin-like receptors KIR.

(26) Another aspect of the invention includes mimetic $\gamma\delta$ T cells comprising cells expressing (a) at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 or a combination thereof; (b) $\gamma\delta$ TCR; and (c) with low or no expression of killer cell immunoglobulin-like receptors KIR.

(27) Other aspects and features of the present invention will become apparent to those of ordinary skill in the art upon review of the following description of specific embodiments of the invention in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the figures, which illustrate, by way of example only, embodiments of the present invention, (2) FIG. 1: Generation of NK cells from hPSCs. (a): A schematic of a two-stage protocol for NK cell derivation from hPSCs. (b-i): Morphological changes during differentiation of H1 cells into NK cells. Phase contrast images show (b): Undifferentiated H1 cells; (c): H1 and OP9 coculture day 12; (d): 1st differentiated cells and OP9-DLL1 coculture day 7 (day 19); (e): 2nd differentiated cells and OP9-DLL1 coculture day 7 (day 26); (f-g): 3rd differentiated cells and OP9-DLL1 coculture day 7 (day 33); (h-i): 4th differentiated cells and OP9-DLL1 coculture day 7 (day 40). (j-l): Phenotypic changes during differentiation of H1 cells into NK cells. Flow cytometric analysis shows U): CD34⁺ cells from H1 and OP9 coculture day 12; (k-l): CD56⁺CD45⁺ cells from 3rd (d33) and 4th (d40) coculture on OP9-DLL1.

(3) FIG. 2: Generation of NK cells from PBC-derived iPSCs. (a) Generation of iPSCs from PBCs. Typical morphology and phenotype of a PBC-derived iPSC (PBC-iPSC) colony and expanded iPSCs. Phase contrast images show a PBC-iPSC colony on day 20 and day 25 and an expanded iPSC line of passage 1 and passage 2. Fluorescence images show the immunostaining of SSEA-4, TRA-1-60, TRA-1-81 and alkaline phosphatase in a PBC-iPSC line. (b-e) Generation of NK cells from PBC-iPSCs. Phenotypic changes during differentiation of a PBC-iPSC line, PBC-iPSC #9 into NK cells are shown. Flow cytometric analysis shows the absence of TCR $\alpha\beta$ expression during the differentiation process. Different cytokines were tested to enhance NK cell commitment. (f). (4) FIG. 3: Generation of $\gamma\delta$ T cells from GDT-iPSCs. A $\gamma\delta$ T cell-derived iPSC line, GDTA/NF-iPSC #1 was differentiated using established protocol in FIG. 1a. (a-l): Phenotype of differentiated cells from GDTA/NF-iPSC #1 line on day 33 and day 47 as analyzed by flow cytometry. (m): Morphology of differentiated cells from GDTA/NF-iPSC #1 line on day 47. A non T cell-derived iPSC line, GDTA/NF-iPSC #3 was differentiated as control. (n-q): Phenotype of differentiated cells from GDTA/NF-iPSC #3 line on day 47 as analyzed by flow cytometry.

(5) FIG. 4: Morphology and phenotype of PBC-iPSC-derived NK cells. (a-b): Morphology (a) and phenotype (b) of NK cells derived from PBC-iPSC #9.

(6) FIG. 5: Expression of activating molecules and death-inducing ligands on iPSC- $\gamma\delta$ T cells, PB- $\gamma\delta$ T cells and PB-NK cells. iPSC- $\gamma\delta$ T cells were generated from GDTA/NNF-iPSC #1 line using established protocol in FIG. 1a. PB- $\gamma\delta$ T cells and PB-NK cells were expanded from PBMCs of a healthy donor. Phenotypes of iPSC- $\gamma\delta$ T cells (a), expanded PB- $\gamma\delta$ T cells (b) and expanded PB-NK cells (c) were analyzed by flow cytometry.

(7) FIG. 6: Expression of Inhibitory receptor and KIRs on iPSC- $\gamma\delta$ T cells, PB- $\gamma\delta$ T cells and PB-NK cells. iPSC- $\gamma\delta$ T cells were generated from GDTA/NF-iPSC #1 line using established protocol in FIG. 1a. PB- $\gamma\delta$ T cells and PB-NK T cells were expanded from PBMCs of a healthy donor. Expression of inhibitory receptor and KIRs on iPSC- $\gamma\delta$ T cells (a), expanded PB- $\gamma\delta$ T cells (b) and expanded PB-NK cells (c) were detected by flow cytometry.

(8) FIG. 7: Functions of PBC-iPSC-derived NK cells. (a-b): IFN- γ secretion by NK cells derived

from PBC-iPSC #9 after stimulation by K562 and Raji cells detected by ELISPOT assay. ELISPOT images (a) and spot counting (b) are shown. (c-d): GrB secretion by NK cells from PBC-iPSC #9 after stimulation by K562 cells detected by ELISPOT assay. ELISPOT images (c) and spot counting (d) are shown. (e-f): Cytotoxicity of NK cells derived from PBC-iPSC #9 against K562 and Raji cells measured by flow cytometry. A representative flow cytometric analysis (e) and a result summary (f) are shown. (g-h): ADCC of NK cells against Raji cells in the presence of anti-CD20 humanized antibody measured by flow cytometry. A representative flow cytometric analysis (g) and a result summary (h) are shown.

(9) FIG. 8: Cytotoxicity of iPSC-derived $\gamma\delta$ NKT cells against cancer cells. $\gamma\delta$ NKT cells were generated from GDTA/NF-iPSC #1 line and used for cytotoxicity assay against a variety of cancer cell lines: MCF7, SW480, BT474 and SK-OV-3.

(10) FIG. 9: Expansion of peripheral blood $\gamma\delta$ T cells. PBMCs were cultured with Zometa and IL-2. (a): Morphology of PBMC culture on day 1 and day 7; (b): Phenotype of PBMC culture on day 7.

(11) FIG. 10: Generation of GDT-iPSCs. Cultured PBMCs were reprogrammed into iPSCs using Sendai viral vectors or nucleofaction. (a-f): Morphological change during reprogramming of cultured PBMCs into iPSCs. Phase contrast images show that (a): cultured PBMCs survived on day 6 after transduction by Sendai viral vectors; (b): an iPSC colony appeared on mEFs on day 17 after seeding the transduced cells; (c): an iPSC colony attached to a Matrigel-coated plate after being transferred from a mEF plate; (d-f): three iPSC lines generated via nucleofaction were established. (g): TCRG gene clonality assay to identify $\gamma\delta$ T cell-derived iPSC lines. The arrow indicates the detection of rearranged TCRG gene in a GDTA/NF-iPSC #1 line. (h): Detection of CD3 and $\gamma\delta$ TCR expression in three established iPSC lines by flow cytometry.

(12) FIG. 11: Expansion of fresh and cryopreserved PBC-iPSC-derived NK cells. (a-b): Expansion of fresh NK cells derived from PBC-iPSC #9 by K562-mb15-41 BBL in G-Rex10 starting with different NK cell numbers. The absolute numbers (a) and fold changes (b) of NK cells during the 14-day expansion are shown. (c-d): Phenotype (c) and cytotoxicity against K562 (d) of fresh NK cells after 14-day expansion measured by flow cytometry. (e): Viability of expanded NK cells after freeze/thaw procedure. (f): Expansion of cryopreserved NK cells. (g): Cytotoxicity of cryopreserved NK cells against K562 after expansion.

DETAILED DESCRIPTION

(13) A robust protocol to consistently generate innate immune cells from various hPSC sources is yet to be established. In this regard an overall manufacturing scheme for mass production of functional NK cells, has been designed and demonstrated not only from a classical hPSC source, human embryonic stem cells (hESCs), but also from a convenient hPSC source, peripheral blood cell (PBC)-derived iPSCs (PBC-iPSCs) without employing CD34⁺ cell enrichment or spin EB formation.

(14) Human pluripotent stem cells (hPSCs), especially induced pluripotent stem cells (iPSCs) provide a promising starting material to produce mimetic innate immune cells such as mimetic natural killer (NK) cells and mimetic $\gamma\delta$ T-cells for cancer immunotherapy. To facilitate consistent mass production, an overall manufacturing scheme to make mimetic innate immune cells from human pluripotent stem cells (hPSCs) was designed and demonstrated. Particularly, a robust protocol to differentiate hPSCs into mimetic NK cells or mimetic $\gamma\delta$ T-cells with NK properties through sequential hematopoietic differentiation on OP9 cells and lymphoid commitment on OP9-DLL1 cells without employing CD34⁺ cell enrichment and spin embryoid body formation.

(15) Accordingly, a first aspect of the invention includes a method of generating mimetic innate immune cells, the method comprising: (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor to generate primed innate immune cells; (b) co-culturing the primed innate immune cells with a stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta

like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(16) In various embodiments using this protocol, functional mimetic NK cells and mimetic $\gamma\delta$ T-cells with NK properties were generated.

(17) As used herein the term 'mimetic innate immune cells' refers to cells that have been derived from pluripotent stem cells and have some similar features of the innate immune cells they mimic and some different features to the innate immune cells that they mimic. In various embodiments the mimetic innate immune cells are called iPSC-derived NK cells. For example, in various embodiments a mimetic innate immune cell such as PBC-iPSC derived NK cell or iPSC-derived $\gamma\delta$ NKT cells have features of NK cells such as expression of NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL and/or NKG2A/CD94 but also new features not present on naturally occurring NK cells or $\gamma\delta$ T cells such as no or low expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cell such as PBC-iPSC-derived NK cell has features of NK cells such as expression of at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 or a combination thereof. These features have the advantage of recognizing a broad spectrum of cancer cells. In this embodiment the cells also have new features such as no or low expression of killer cell immunoglobulin-like receptors (KIRs). In various other embodiments the mimetic innate immune cell such iPSC-derived $\gamma\delta$ NKT cell expresses $\gamma\delta$ TCRs that can recognize phosphoantigen besides having any one of the above-mentioned features present in the PBC-iPSC-derived NK cell. In various embodiments low KIR refers to less than 20% expression of KIR; or less than 10% expression of KIR; or less than 5% expression of KIR. In various embodiments the mimetic innate immune cell have 5% or less expression of KIR2DL1. In various embodiments the mimetic innate immune cell have 13% or less expression of KIR3DL1. In various embodiments the mimetic innate immune cells have 7% or less expression of KIR3DL1. In various embodiments the mimetic innate immune cells have 18% or less expression of KIR2DL2. A KIR-negative or KIR-low expression mimetic innate immune cell has the advantage that it may serve a wide range of recipients disregarding their HLA genotypes. Interestingly, starting with an autologous PBC-iPSC line, it is possible to generate autologous KIR-negative or KIR-low expression mimetic innate immune cell source to be used under an autologous setting without worrying about the inhibition imposed by self HLA molecules. With a reduced risk of immune rejection, these mimetic innate immune cells such as autologous PBC-iPSC-NK cells may survive longer and thus provide a prolonged anti-tumor activity.

(18) In various embodiments the mimetic innate immune cell such as iPSC-derived $\gamma\delta$ NKT cells have features of NK cells such as expression of at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94, or a combination thereof and new features such as no or low expression of killer cell immunoglobulin-like receptors (KIR) but also features of $\gamma\delta$ T cells such as CD3⁺ and/or $\gamma\delta$ T cell receptors such as V gamma 9V delta 2 (V γ 9V δ 2).

(19) A stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) may either have a deficient gene that is either not expressed or expresses a non-functional mutant or it may be knocked down using any known method such as antisense, siRNA, shRNA or any other method known in the art to knock down protein expression. In various embodiments the stromal cell line deficient in expressing M-CSF may be a mesenchymal stem cell such as a bone marrow mesenchymal stem cell. In various embodiments the stromal cell line deficient in expressing M-CSF may be an OP9 cell. In various embodiments the M-CSF functional cDNA is represented by nucleic acid sequence SEQ ID NO. 2, and a functional M-CSF protein is represented by amino acid sequence SEQ ID NO. 3. Any mutation to these sequences that results in a dysfunctional M-CSF

would be sufficient.

(20) In various embodiments to develop an industry-compatible differentiation protocol, it has been demonstrated that by co-culturing with OP9, a classical M-CSF-deficient stromal cell line, efficient hematopoietic differentiation of both hESCs and PBC-iPSCs can be consistently achieved in 12 days, which is significantly shorter than using other stromal cell lines. The harvested hematopoietic cells can be directly used for NK cell generation without sorting CD34⁺ cells. Thus, using OP9 to induce hematopoietic differentiation can significantly simplify the manufacturing process by avoiding CD34⁺ cell sorting and EB formation. In various embodiments a progenitor of innate immune cell comprises a hematopoietic cell.

(21) A stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) and ectopically expressing Notch ligand, Delta like 1 (DLL1) may be a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) as discussed above that has been engineered to express DLL1 exogenously or abnormally. Any method known in the art to artificially cause expression of DLL1 in the stromal cell line deficient in expressing M-CSF would be suitable. In various embodiments the stromal cell line deficient in expressing M-CSF ectopically expressing DLL1 may be a mesenchymal stem cell such as a bone marrow mesenchymal stem cell transduced with DLL1. In various embodiments the stromal cell line deficient in expressing M-CSF ectopically expressing DLL1 may be an OP9 DLL1 cell. In various embodiments the DLL1 cDNA is represented by nucleic acid sequence SEQ ID NO. 4, and a DII1 protein is represented by amino acid sequence SEQ ID NO. 5. Functional variations of the sequences that result in a functional DLL1 would be sufficient.

(22) In various embodiments to provide a robust microenvironment for NK cell development, an OP9 cell line modified with DLL1 together with an NK cell-promoting cytokine combination including SCF, FLT3L, IL-7 and IL-15 was used. Using such an approach, the heterogeneous hematopoietic cells harvested from hPSC/OP9 co-cultures can be directly used for NK cell generation and the end product is a homogenous lymphoid population with more than 99% CD56⁺CD45⁺NK cells. The robustness of such method is likely due to the activation of Notch signaling pathway, which plays an important role in the development of innate lymphoid cells. This differentiation approach excludes the need for cell sorting, EB formation, spin EB formation, T or B cell depletion or NK cell enrichment.

(23) Without being limited to any theories it is postulated that the deficiency of the macrophage colony stimulating factor (M-CSF) in step (a) sequentially differentiates the cells to hematopoietic cells and the combination of the deficiency of the macrophage colony stimulating factor (M-CSF) together with expression of DLL1 in the media in step (b) stimulates lymphoid commitment and this two phase process combination of the two different types of stromal cells used one after the other results in the mimetic innate immune cells with unique features. In various embodiments the process facilitates the expression of features of NK cells such as expression of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL NKG2A/CD94 or a combination thereof as well as other features not common to NK cells such as no or low expression of killer cell immunoglobulin-like receptors (KIR).

(24) In various embodiment the media used in step (b) also plays a large role and in various embodiments the media comprises stem cell factor (SCF) Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7). In various embodiments the media comprises stem cell factor (SCF) Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 15 (IL15). In various other embodiments the media comprises stem cell factor (SCF) Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and interleukin 15 (IL15). While all three combinations are able to generate the mimetic innate immune cells with unique features. Surprisingly, the use of interleukin 15 (IL15) alone or in combination with interleukin 7 (IL7) significantly increased the percentage of mimetic innate immune cells generated although the use of interleukin 7 (IL7) alone still successfully generated the mimetic innate immune cells with unique features. Other lymphocyte culture medium

known in the art can also be used as basal medium.

(25) As used herein the term 'passage', 'passaging' or 'passaged' refers to sub-culturing some or all of the cells into fresh media. This may involve any of the methods commonly known or used in the art. In various embodiments the differentiated mimetic innate immune cells are passaged after every 7 days of co-culturing 3 times and the final mimetic innate immune cells are harvested after 21 days, preferably by 33 days. In various embodiments the differentiated mimetic innate immune cells are passaged after every 7 days of co-culturing 4 times and the final mimetic innate immune cells are harvested after 28 days, preferably by 40 days. In various embodiments the differentiated mimetic innate immune cells are passaged after every 7 days of co-culturing 5 times and the final mimetic innate immune cells are harvested after 35 days, preferably by 47 days.

(26) In various embodiments the human pluripotent stem cells in step (a) are co-cultured for 12 days with the stromal cell line deficient in expressing M-CSF. In various embodiments human pluripotent stem cells in step (a) are co-cultured in a simple media comprising alpha Minimum Essential Medium (α MEM) and Fetal Bovine Serum (FBS) and the co-cultures were fed every 3-4 days by changing half the simple media. Other stem cell differentiation media known in the art may also be used.

(27) In various embodiments the media used in step (b) further comprises Fetal Bovine Serum (FBS); alpha Minimum Essential Medium (α MEM); or a combination thereof.

(28) In various embodiments the human pluripotent stem cells are human embryonic stem cells (hESC). hESCs are a good option in terms of safety, but the derivation of hESCs is always ethically controversial and the applications of hESC derivatives are limited to allogeneic settings.

(29) In various embodiments the human pluripotent stem cells are induced from peripheral blood cells (PBC (iPSC)). Using iPSC has the advantage that they can be generated from a reliable, unlimited and standardizable starting cell source such as peripheral blood mononuclear cells (PBMC) isolated from donors or patients allowing iPSCs derivatives to be used in both autologous and allogeneic applications and it further doesn't have any of the ethical concerns of using hESC

(30) Currently, most iPSCs are generated by reprogramming adult somatic cells. The choice of starting somatic cells will affect not only the efficiency and kinetic of reprogramming, but also the practicality of generating GMP-grade iPSCs. Although fibroblasts are the most commonly used somatic cells, they are not very GMP-compliant. Skin sample collection through punch biopsy is invasive and growing fibroblasts from skin biopsy sample is time-consuming (up to 3 weeks). The derivation of fibroblasts under GMP itself is already a daunting task. In contrast, using PBCs to generate iPSCs is a more practical option since peripheral blood collection is convenient and isolation of mononuclear cells from peripheral blood sample only takes 15 minutes. The easiness of implementing GMP in sample collection, transportation and processing renders PBCs an attractive starting material for iPSC derivation. Most importantly, using the two-stage differentiation protocol, for the first time the generation of high-purity, functional and expandable mimetic innate immune cells such as NK cells from PBC-derived iPSCs was demonstrated.

(31) In various embodiments reprogramming transcription factors are incorporated into the peripheral blood cells and grown to generate induced pluripotent stem cells (iPSC).

(32) In various embodiments the reprogramming transcription factors include the traditional Yamanaka transcription factors Oct 4, Sox2, Klf4 and c-myc. In various other embodiments the reprogramming transcription factors may be Oct 3/4 and a member of the Sox family including any one of Sox 1, Sox 2, Sox 3 or Sox 15. Alternatively, in various other embodiments the reprogramming transcription factors may be Oct 3/4; a member of the Sox family of transcription factors including any one of Sox 1, Sox 2, Sox 3 or Sox 15; NANOG and LIN28. Alternatively, in various other embodiments the reprogramming transcription factors may be Oct 3/4; a member of the Sox family of transcription factors including any one of Sox 1, Sox 2, Sox 3 or Sox 15; a member of the Klf family of transcription factors including Klf1, Klf 2, Klf4 or Klf5 and a member of the Myc family of transcription factors including c-myc, L-myc or N-myc. Alternatively, in

various other embodiments the reprogramming transcription factors may be Oct 3/4; a member of the Sox family of transcription factors including any one of Sox 1, Sox 2, Sox 3 or Sox 15; a member of the Klf family of transcription factors including Klf1, Klf 2, Klf4 or Klf5 and Glis1 or any reprogramming transcription factors known in the art to effectively generate pluripotent cells that express stage-specific embryonic antigens (SSEA-3, and SSEA-4), and/or epitopes recognized by TRA-1-60, TRA-1-81 antibodies, initiated from a differentiated adult cell.

(33) In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by retroviral transduction. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by lentiviral expression system. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood using a single cassette reprogramming vector in which each of the reprogramming factors are separated by a self-cleaving peptide signal such as STEMCCA. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by adenovirus expression system. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by sendai virus expression system. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by diffusion of the proteins into the cell. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by mRNA transfection. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by miRNA infection or transfection. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by a PiggyBac system that exploits the use of a transposon system. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by minicircle vectors containing only a eukaryotic promotor and the cDNA to be expressed. In various embodiments the reprogramming transcription factors may be incorporated into the peripheral blood cells by transient expression of the reprogramming factors in an episomal plasmid or any other method known in the art to deliver the reprogramming transcription factors into the peripheral blood cells to efficiently generate an induced pluripotent stem cells (iPSC).

(34) In various embodiments the method further comprises expanding peripheral blood cells in the presence of an amino-bisphosphonate and interleukin 2 (IL2) prior to incorporating reprogramming transcription factors into the peripheral blood cells to generate induced pluripotent stem cells (iPSC).

(35) In various embodiments amino-bisphosphonate and interleukin 2 (IL2) are used to induce the expansion of $\gamma\delta$ T cells generally that are found to express V γ 9V δ 2 T cells. Without being limited to any theories it is postulated that as V γ 9V δ 2 T cells reacts to phosphoantigens the amino-bisphosphonate stimulates the formation of such V γ 9V δ 2 T cells.

(36) In various embodiments the amino-bisphosphonate is zoledronate or zoledronic acid or salts thereof commercially known as Zometa. Preferably because zoledronic acid is easily obtainable commercially however any amino-bisphosphonate would be suitable. The formula for zoledronate is as follows.

(37) ##STR00001##

(38) In various embodiments the iPSC comprises a TCR gene arranged in the TCRG gene configuration. In various embodiments a TCRG gene configuration comprises the somatic recombination of TCR to TCRG and may also include TCRD gene configuration comprising the somatic recombination of TCR to TCRD. In various embodiments the TCRG gene configuration is represented by nucleic acid sequence SEQ ID NO. 1, functional variations of the sequence that result in the expression of a functional TCR γ would also be included. In various embodiments the iPSC comprises a TCR gene arranged in the TCRD gene configuration. In various embodiments a TCRD gene configuration comprises the somatic recombination of TCR to TCRD of any known TCRD sequence that results in the expression of a functional TCR δ would be suitable.

(39) As used herein a TCR gene arranged in the TCRG gene configuration comprises a nucleic acid molecule which molecule typically encodes a γ TCR polypeptide, allelic variant, or analog, including fragments, thereof. Specifically provided are DNA molecules for use in screening for the TCRG gene configuration and DNA molecules for securing expression of a γ TCR polypeptide selected from the group consisting of: (a) DNA molecules set out in SEQ ID NO: 1, or fragments thereof; (b) DNA molecules that hybridize to the DNA molecules defined in (a) or hybridisable fragments thereof; and (c) DNA molecules that code on expression for the amino acid sequence encoded by any of the foregoing DNA molecules.

(40) A polynucleotide is said to “encode” a polypeptide if, in its native state or when manipulated by methods well known to those skilled in the art, it can be transcribed and/or translated to produce the mRNA for and/or the polypeptide or a fragment thereof. The anti-sense strand is the complement of such a nucleic acid, and the encoding sequence can be deduced therefrom.

(41) “TCRG gene sequence,” “TCRG gene,” “TCRG nucleic acids” or “TCRG polynucleotide” each refer to polynucleotides that are likely to be expressed in normal T-cells. Mutations at the TCRG gene sequence that are capable of expressing a functional γ TCR polypeptide are included.

(42) The TCRG gene sequence is intended to include coding sequences, intervening sequences and regulatory elements controlling transcription and/or translation. The TCRG gene sequence is intended to include all allelic variations of the DNA sequence that are capable of expressing a γ TCR polypeptide.

(43) A nucleic acid or fragment thereof is “substantially homologous” (“or substantially similar”) to another if, when optimally aligned (with appropriate nucleotide insertions or deletions) with the other nucleic acid (or its complementary strand), there is nucleotide sequence identity in at least about 60% of the nucleotide bases, usually at least about 70%, more usually at least about 80%, preferably at least about 90%, and more preferably at least about 95-98% of the nucleotide bases.

(44) Other preferred polynucleotides comprise a contiguous sequence having greater than 40, 50, 60, or 70% identity, more preferably greater than 80, 90, 95 or 97% identity to the sequence of SEQ ID NO: 1 that encodes amino acids for γ TCR.

(45) Detection of the TCRG gene configuration may be done in accordance with methods known in the art including techniques such as Southern blot hybridisation, polymerase chain reaction, or other suitable methods may be used.

(46) In various embodiments this has the advantage of being the first demonstration of using $\gamma\delta$ T cell-derived iPSCs to generate $\gamma\delta$ T cells. An industry-compatible production scheme and method to derive iPSCs from $\gamma\delta$ T cells and use these iPSCs to generate $\gamma\delta$ T cells and a novel immune cell type— $\gamma\delta$ natural killer T cells is derived.

(47) In various embodiments the method further comprises expanding mimetic innate immune cells in the presence of feeder cells in a media comprising stem cell factor (SCF); Fetal Bovine Serum (FBS); Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 15 (IL15) and interleukin 7 (IL7).

(48) In various embodiments the feeder cells are irradiated cell line able to expand NK cells. In various embodiments the feeder cells are gamma irradiated K562-mbIL15-41BBL.

(49) In various embodiments the PBC-iPSC-derived mimetic NK cells could be expanded by 74-fold in 9 days and by 38.5-fold after cryopreservation using a feeder cell line. These expanded mimetic NK cells became more potent functionally as shown by the increased cytotoxicity against K562 cells. The possibility to generate iPSCs from both fresh and frozen PBCs provides even more flexibility for GMP manufacturing. To this end, the feasibility to derive iPSCs from frozen PBCs was demonstrated in this study.

(50) Another aspect of the invention includes mimetic innate immune cells comprise cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 or a combination thereof, and with low or no expression of killer cell immunoglobulin-like receptors (KIR). These mimetic innate immune cells are optionally cells with or without $\gamma\delta$ TCR expression.

(51) In various embodiments the mimetic innate immune cells comprise cells expressing CD56, and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing CD16, and with low or no expression of killer cell immunoglobulin-like receptors (KIR) as described above herein. In various embodiments the mimetic innate immune cells comprise cells expressing NKp30 and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing NKp44 and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing NKp46 and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing NKG2D and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing DNAM-1 and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing FASL and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing TRAIL and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing NKG2A/CD94 and with low or no expression of killer cell immunoglobulin-like receptors (KIR). In various embodiments the mimetic innate immune cells comprise cells expressing a combination of any 2, or any 3, or any 4, or any 5, or any 6, or any 7, or any 8, or any 9, or all of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 and with low or no expression of killer cell immunoglobulin-like receptors (KIR).

(52) In various embodiments cell expression profiles may be determined by any means known in the art for measuring protein expression. For example chromatography methods such HPLC, LCMS, or polyacrylamide gel electrophoresis. Alternatively antibody methods where an antibody that hybridizes to the protein is tagged for detection such as ELISA, flow cytometry, protein immunoprecipitation, immune-electrophoresis, or western blotting. In various embodiments an antibody with a conjugated tag, such as a colour tag, a magnetic tag, a fluorescent tag, a radioactive tag, a protein tag with known properties, is able to hybridise to a portion of any one of the protein sequences CD56 represented by amino acid sequence SEQ ID NO: 6, CD16 represented by amino acid sequence SEQ ID NO: 7, NKp30 represented by amino acid sequence SEQ ID NO: 8, NKp44 represented by amino acid sequence SEQ ID NO: 9, NKp46 represented by amino acid sequence SEQ ID NO: 10, NKG2D represented by amino acid sequence SEQ ID NO: 11, DNAM-1 represented by amino acid sequence SEQ ID NO: 12, FASL represented by amino acid sequence SEQ ID NO: 13, TRAIL represented by amino acid sequence SEQ ID NO: 14, or NKG2A/CD94 represented by amino acid sequence SEQ ID NO: 15.

(53) In various embodiments the mimetic innate immune cells are cells with or without $\gamma\delta$ TCR expression.

(54) In various embodiments the mimetic innate immune cells are as described herein.

(55) In various embodiments the mimetic innate immune cells further comprise expression of CD3+ a feature of some T-cells.

(56) In various embodiments the mimetic innate immune cells further comprise expression of gamma delta TCR a feature of gamma/delta T-cells.

(57) In various embodiments the gamma delta TCR comprises a V gamma 9V delta 2 (V γ 9V δ 2).

(58) In various embodiments the technology has demonstrated a best possible method to generate $\gamma\delta$ T cells and a novel immune cell type— $\gamma\delta$ natural killer T cells from iPSCs.

(59) In various embodiments to develop $\gamma\delta$ T cells into unlimited “off-the-shelf” products, it was demonstrated for the first time the generation of $\gamma\delta$ T cells from induced pluripotent stem cells (iPSCs). Starting with $\gamma\delta$ T cell-enriched blood cell cultures, iPSC lines were generated and

identified as those derived from $\gamma\delta$ T cells. In various embodiment using these $\gamma\delta$ T cell-derived iPSCs (GDT-iPSCs), $\gamma\delta$ T cells were generated, which were $\gamma\delta$ TCR+CD3+ and V γ 9V δ 2+. Moreover, by differentiating GDT-iPSCs with the herein disclosed protocol for generating NK cells from iPSCs, the generated $\gamma\delta$ T cells were further armed with many “cancer-fighting” molecules of NK cells, which include activating receptors, natural cytotoxicity receptors and death-inducing molecules. Thus, equipped with anticancer features of both $\gamma\delta$ T cells and NK cells, these novel iPSC-derived “ $\gamma\delta$ natural killer T cells” have great therapeutic activity to target a broad range of cancer cells.

(60) Another aspect of the invention includes mimetic innate immune cells as described herein for use in a treatment.

(61) In various embodiments the mimetic innate immune cells may be suitable for treating viral infections, bacterial infections, fungal infections any other infections or cancer.

(62) Another aspect of the invention includes mimetic innate immune cells as described herein for use in treating cancer.

(63) In various embodiments the mimetic innate immune cells may be suitable for treating cancer including sarcomas, carcinomas, and lymphomas, leukemia, breast cancer, colorectal cancer, or ovarian cancer.

(64) In various embodiments the mimetic innate immune cells may be suitable for treating solid tumor cancer. In various embodiments the solid tumor includes sarcomas, carcinomas, or lymphomas.

(65) In various embodiments the mimetic innate immune cells may be suitable for treating breast adenocarcinoma, colorectal adenocarcinoma, breast ductal carcinoma, ovary adenocarcinoma breast ductal carcinoma, ovary adenocarcinoma or leukemia.

(66) In various embodiments leukemia comprises chronic myelogenous leukemia or Burkitt's lymphoma.

(67) In various embodiments the mimetic innate immune cells were shown to be toxic against cancer cells such as human myelogenous leukemia cells, breast adenocarcinoma cells, colorectal adenocarcinoma cells, breast ductal carcinoma cells, and ovary adenocarcinoma cells.

(68) Another aspect of the invention includes use of mimetic innate immune cells as described herein for manufacture of a medicament in the treatment of cancer.

(69) In various embodiments the cancer is a solid tumor cancer or leukemia.

(70) In various embodiments the solid tumor cancer comprises breast adenocarcinoma, colorectal adenocarcinoma, breast ductal carcinoma, or ovary adenocarcinoma. In various embodiments the leukemia comprises acute lymphoblastic leukemia, acute myeloid leukemia, chronic lymphocytic leukemia or chronic myeloid leukemia.

(71) In various embodiments cancer includes sarcomas, carcinomas, lymphomas, leukemia, breast cancer, colorectal cancer, or ovarian cancer.

(72) The mimetic innate immune cells were shown to be toxic against cancer cells such as human myelogenous leukemia cells, breast adenocarcinoma cells, colorectal adenocarcinoma cells, breast ductal carcinoma cells, and ovary adenocarcinoma cells.

(73) Another aspect of the invention includes a kit for generating mimetic innate immune cells, the kit comprising: (a) a human pluripotent stem cell line; (b) a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF); (c) a stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta like 1 (DLL1); (d) a first simple media for co-culturing a human pluripotent stem cells with the stromal cell line deficient in expressing M-CSF; and (e) a second media comprising Stem cell Factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15).

(74) In various embodiments the human pluripotent stem cell line may include a human embryonic stem cell (hESC). In various other embodiments the human pluripotent stem cell line may include a

Peripheral Blood Cell-derived iPSC line that may be a non-T cell-derived iPSC line or a $\gamma\delta$ cell-derived iPSC line.

(75) The stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) and the stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta like 1 (DLL1) are as described herein above.

(76) In various embodiments the first simple media comprises alpha Minimum Essential Medium (α MEM) or other stem cell differentiation medium and Fetal Bovine Serum (FBS).

(77) In various embodiments the second media comprises Stem cell Factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7). In various embodiments the media comprises Stem cell Factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), Interleukin 15 (IL15). In various other embodiments the media comprises Stem cell Factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and interleukin 15 (IL15). While all three combinations are able to generate the mimetic innate immune cells with unique features.

Surprisingly, the use of interleukin 15 (IL15) alone or in combination with interleukin 7 (IL7) significantly increased the percentage and/or amount of mimetic innate immune cells generated although the use of interleukin 7 (IL7) alone still successfully generated the mimetic innate immune cells with unique features.

(78) In various embodiments the second media further comprises Fetal Bovine Serum (FBS); alpha Minimum Essential Medium (α MEM) or other lymphocyte culture medium; or a combination thereof.

(79) In various embodiments the kit further comprises an amino-bisphosphonate and interleukin 2 (IL2). In various embodiments the amino-bisphosphonate is zoledronic acid or salts thereof.

(80) Another aspect of the invention includes a method of treating a patient with cancer comprising: administering to the patient mimetic innate immune cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL NKG2A/CD94 or a combination thereof and low or no expression of killer cell immunoglobulin-like receptors KIR generated by (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor to generate progenitors of innate immune cells; (b) co-culturing the progenitors of innate immune cells with a stromal cell line deficient in expressing macrophage colony stimulating factor and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(81) In various embodiments the cancer is a solid tumor cancer as described herein. In various embodiments the cancer is a leukemia as described herein.

(82) In various embodiments the solid tumor cancer comprises breast adenocarcinoma, colorectal adenocarcinoma, breast ductal carcinoma, or ovary adenocarcinoma. In various embodiments the leukemia comprises acute lymphoblastic leukemia, acute myeloid leukemia, chronic lymphocytic leukemia or chronic myeloid leukemia.

(83) Another aspect of the invention includes a method of generating induced pluripotent stem cells (iPSC) from peripheral blood mononuclear cells, comprising: (a) incorporating reprogramming transcription factors into peripheral blood mononuclear cells; (b) growing the peripheral blood mononuclear cells incorporated with the reprogramming transcription factors into pluripotent stem cells.

(84) In various embodiments the mononuclear peripheral blood cells are $\gamma\delta$ T cells. This is the first demonstration of generating iPSCs from $\gamma\delta$ T-cells. In various other embodiments the mononuclear peripheral blood cells are non-T cells of peripheral blood cells.

(85) In various embodiments the method further comprises expanding peripheral blood cells in the presence of an amino-bisphosphonate and interleukin 2 (IL2) prior to incorporating reprogramming transcription factors into the gamma delta T-cells.

(86) In various embodiments the amino-bisphosphonate is zoledronic acid or salts thereof.

(87) In various embodiments the iPSC comprises a TCR gene arranged in the TCRG gene configuration or a TCR gene arranged in the TCRD gene configuration. In various embodiments the TCRG gene configuration is represented by nucleic acid sequence SEQ ID NO. 1. Functional variations of the sequence that result in the expression of a functional TCR γ would also be sufficient. In various embodiments a TCRD gene configuration comprises the somatic recombination of TCR to TCRD of any known TCRD sequence that results in the expression of a functional TCR δ would be suitable.

(88) Another aspect of the invention includes an induced pluripotent stem cells (iPSC) comprising a TCR gene arranged in the TCRG gene configuration or a TCR gene arranged in the TCRD gene configuration.

(89) In various embodiments the rearranged TCRG gene is represented by nucleic acid sequence SEQ ID NO. 1. Functional variations of the sequences that result in the expression of a functional TCR γ would be sufficient. In various embodiments a TCRD gene configuration comprises the somatic recombination of TCR to TCRD of any known TCRD sequence that results in the expression of a functional TCR δ would be suitable.

(90) Another aspect of the invention includes a method of generating mimetic NK cells, the method comprising: (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) to generate progenitors of innate immune cells; (b) co-culturing the progenitors of innate immune cells with a stromal cell line deficient in expressing M-CSF and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(91) Another aspect of the invention includes a method of generating mimetic $\gamma\delta$ cells, the method comprising: (a) co-culturing human pluripotent stem cells with a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) to generate primed innate immune cells; (b) co-culturing the primed innate immune cells with a stromal cell line deficient in expressing M-CSF and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and/or interleukin 15 (IL15) to generate differentiated mimetic innate immune cells; (c) passaging the differentiated mimetic innate immune cells weekly for 3 to 5 weeks.

(92) Another aspect of the invention includes mimetic NK cells comprising cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 or a combination thereof and with low or no expression of killer cell immunoglobulin-like receptors KIR.

(93) In various embodiments the mimetic NK cells comprise cells expressing no CD3 or no $\gamma\delta$ TCR.

(94) Another aspect of the invention includes mimetic gamma delta T-cells comprising cells expressing at least one of CD56, CD16, NKp30, NKp44, NKp46, NKG2D, DNAM-1, FASL, TRAIL, NKG2A/CD94 or a combination thereof and with low or no expression of killer cell immunoglobulin-like receptors KIR.

(95) In various embodiments the mimetic $\gamma\delta$ T cells are cells with $\gamma\delta$ TCR expression or CD3 expression.

(96) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by a skilled person to which the subject matter herein belongs. As used herein, the following definitions are supplied in order to facilitate the understanding of the present invention.

(97) Throughout this document, unless otherwise indicated to the contrary, the terms “comprising”, “consisting of”, “having” and the like, are to be construed as non-exhaustive, or in other words, as

meaning “including, but not limited to”.

(98) Furthermore, throughout the specification, unless the contest requires otherwise, the word “include” or variations such as “includes” or “including” will be understood to imply the inclusion of a stated integer or group of integers but not the exclusion of any other integer or group of integers.

(99) As used in the specification and the appended claims, the singular form “a”, and “the” include plural references unless the context clearly dictates otherwise.

EXAMPLES

(100) Described is an overall scheme for mass production of innate immune cells from hPSCs. Comparing with the existing protocols, the approach is more industry-compliant and provides the conveniences required to produce NK cell therapeutics: (1) the starting material PBC-iPSCs are a convenient, highly accessible and GMP-compatible hPSC source; (2) the differentiation process is simplified by excluding cell sorting, EB formation and spin EB formation; (3) the generated PBC-iPSC-derived cells are already a high-purity and functional population, obviating the need for T-cell and B-cell depletion or innate immune cell enrichment; (4) both fresh and cryopreserved NK cells can be expanded in a short period of time, which eases the logistics of transporting and manufacturing of NK cell therapeutics; (5) further functional maturation can be achieved after expansion; (6) most of the PBC-iPSC-derived innate immune cells are KIR-negative, which means that these innate immune cells may serve as universal cell source for various recipients without concerning the inhibitory signaling that may reduce the cytotoxicity. Thus, using PBC-iPSCs to derive innate immune cells is an amenable platform for mass production of innate immune cell therapeutics.

(101) Generation of Mimetic Innate Immune Cells

(102) A robust protocol to consistently generate innate immune cells from various hPSC sources is yet to be established. In this regard an overall manufacturing scheme for mass production of functional NK cells, has been designed and demonstrated not only from a classical hPSC source, human embryonic stem cells (hESCs), but also from a convenient hPSC source, peripheral blood cell (PBC)-derived iPSCs (PBC-iPSCs) without employing CD34+ cell enrichment or spin EB formation.

(103) Generation of NK Cells from hESCs.

(104) Cell Culture.

(105) An hPSC line, H1 (WiCell Research Institute, Madison, WI, <http://www.wicell.org>), was cultured with mTeSR1 (StemCell Technologies, Vancouver, BC, Canada, <http://www.stemcell.com>) on Matrigel (BD Biosciences, San Diego, CA, <http://www.bdbiosciences.com>)—coated six-well plates. OP9 cells (American Type Culture Collection [ATCC], Manassas, VA, <http://www.atcc.org>) and OP9/G-DLL1 cells (Riken BRC Cell Bank, Ibaraki, Japan, <http://cell.brc.riken.jp/en/>) were cultured with α -MEM (Thermo Fisher Scientific, Waltham, MA, <http://corporate.thermofisher.com>) supplemented with 20% fetal bovine serum (FBS) (HyClone, Logan, UT, <http://www.hyclone.com>).

(106) Two-Stage Protocol.

(107) In the first stage, OP9 cells were seeded on 0.1% gelatin (StemCell Technologies)—coated T75 flask. Upon confluence, the OP9 cell cultures were fed by changing half of the medium and overgrown for 4-6 days. $1-1.5 \times 10^6$ hESCs were then seeded and differentiated on the overgrown OP9 cells in α -MEM supplemented with 10% FBS for 12 days. The hESC/OP9 co-cultures were fed every 4 days by changing half of the medium. In the second stage, the differentiated cells were harvested from the hESC/OP9 co-cultures using 1 mg/ml collagenase IV (StemCell Technologies) and TrypLE Express (Thermo Fisher Scientific). OP9 cells were removed by plastic adherence for 45 minutes and the cell clumps were further removed by 100 μ m cell strainers (BD Biosciences). The remaining non-adherent cells were then co-cultured with OP9-DLL1 grown on T75 flasks using α -MEM containing 10% FBS, 10 ng/ml SCF (PeproTech, Rocky

Hill, NJ, <http://www.peprotech.com>), 5 ng/ml FLT3L (PeproTech) together with 5 ng/ml IL-7 (PeproTech) and/or 10 ng/ml IL-15 (PeproTech) for 7 days. Hereafter, the differentiated cells were harvested using Versene (Thermo Fisher Scientific) and co-cultured on new OP9-DLL1 grown on six-well plates on a weekly basis for another 3-4 weeks.

(108) To establish a robust two-stage protocol for NK cell derivation from hPSCs (FIG. 1a), a widely used hESC line, H1 (FIG. 1b) was used. In the first stage, to induce hematopoietic differentiation, H1 cells were co-cultured with overgrown OP9 cells, a bone marrow stromal cell line. On day 12 of co-culture, a lot of differentiated colonies (FIG. 1c) and a small population of CD34⁺ cells (FIG. 1j) were consistently observed. In the second stage, to induce lymphoid commitment, these differentiated cells were harvested and co-cultured with OP9-DLL1, an OP9 cell line modified with a Notch ligand Delta-like-1 (DLL1), on weekly basis in the presence of stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L) and IL-7 (FIG. 1a). Seven days after first co-culture with OP9-DLL1 (day 19), most differentiated cells still grew as adherent cells (FIG. 1d); after second co-culture (day 26), some bright round and semi-attached cells started to pop out (FIG. 1e); after third co-culture (day 33), these cells detached and assembled in the center of the well (FIGS. 1f, g); after fourth co-culture (day 40), more suspension cells appeared (FIGS. 1h, i). Morphologically, these suspension cells were small, round and bright lymphoid cells (FIGS. 1g, i); phenotypically, they were a defined population on the scatter plots (FIGS. 1k, l); most of them were CD45^{sup.}+CD56^{sup.}+, but CD3 TCR $\alpha\beta$ ^{sup.}-CD4^{sup.}-CD8^{sup.}- (FIGS. 1k, l), which is typical for NK cells. Thus, by going through hematopoietic differentiation and lymphoid commitment, NK cells can be derived from hESCs without enriching CD34⁺ cells or forming spin EBs.

(109) Generation of NK Cells from iPSCs

(110) Cell Culture.

(111) OP9 cells (American Type Culture Collection [ATCC], Manassas, VA, <http://www.atcc.org>) and OP9/G-DLL1 cells (Riken BRC Cell Bank, Ibaraki, Japan, <http://cell.brc.riken.jp/en/>) were cultured with α -MEM (Thermo Fisher Scientific, Waltham, MA, <http://corporate.thermofisher.com>) supplemented with 20% fetal bovine serum (FBS) (HyClone, Logan, UT, <http://www.hyclone.com>).

(112) Two-Stage Protocol.

(113) In the first stage, OP9 cells were seeded on 0.1% gelatin (StemCell Technologies)—coated T75 flask. Upon confluence, the OP9 cell cultures were fed by changing half of the medium and overgrown for 4-6 days. $1-1.5 \times 10^6$ hESCs were then seeded and differentiated on the overgrown OP9 cells in α -MEM supplemented with 10% FBS for 12 days. The hESC/OP9 co-cultures were fed every 4 days by changing half of the medium. In the second stage, the differentiated cells were harvested from the hESC/OP9 co-cultures using 1 mg/ml collagenase IV (StemCell Technologies) and TrypLE Express (Thermo Fisher Scientific). OP9 cells were removed by plastic adherence for 45 minutes and the cell clumps were further removed by 100 μ m cell strainers (BD Biosciences). The remaining non-adherent cells were then co-cultured with OP9-DLL1 grown on T75 flasks using α -MEM containing 10% FBS, 10 ng/ml SCF (PeproTech, Rocky Hill, NJ, <http://www.peprotech.com>), 5 ng/ml FLT3L (PeproTech) together with 5 ng/ml IL-7 (PeproTech) and/or 10 ng/ml IL-15 (PeproTech) for 7 days. Hereafter, the differentiated cells were harvested using Versene (Thermo Fisher Scientific) and co-cultured on new OP9-DLL1 grown on six-well plates on a weekly basis for another 3-4 weeks.

(114) Generation of NK Cells from PBC-iPSCs.

(115) To further facilitate the production of NK cells from hPSCs under Good Manufacturing Practices (GMP), it is important to start with a convenient hPSC source that is GMP-compatible. Although hESCs and iPSCs generated from fibroblasts or umbilical cord blood CD34⁺ cells have been used to derive NK cells, iPSCs generated from peripheral blood cells (PBC-iPSCs) are a novel, but more accessible and practicable option from the standpoint of manufacturing. To

demonstrate the feasibility of using PBC-iPSCs for NK cell generation, iPSC lines from PBCs using Sendai viral vectors carrying the reprogramming factors were first generated (FIG. 2a) and described below. These resulting PBC-iPSCs showed typical hPSC morphology and phenotype (FIG. 2a).

(116) Next, the two-stage protocol for NK cell generation was examined to determine whether the method is still applicable to these PBC-iPSCs. Using a PBC-iPSC #9 line (FIG. 2b), differentiated cells containing a significant CD34⁺ population were derived after 12-day co-culture with OP9 (FIG. 2c). Without sorting the CD34⁺ cells, these differentiated cells were harvested and directly co-cultured with OP9-DLL1. A small CD45⁺CD56⁺ population appeared after second co-culture (day 26) (FIG. 2d); this population became much more apparent after fifth co-culture (day 47), which comprised 61% of the lymphoid cells (FIG. 2e). Neither TCR $\alpha\beta$ nor CD3 expression was observed throughout the differentiation process: not in the pluripotent stem cells (FIG. 2b), hematopoietic cells (FIG. 2c) or lymphoid cells (FIG. 2d, e). Moreover, there was no obvious CD4 or CD8 expression in the lymphoid cells (FIG. 2d, e), confirming that the two-stage protocol can also be used for NK cell generation from PBC-iPSCs.

(117) Improving Purity and Yield of NK Cells Derived from PBC-iPSCs.

(118) In addition to a GMP-compatible starting hPSC source, further improvement in purity and yield of NK cells derived from PBC-iPSCs will facilitate the production and translational use of this novel NK cell source. To this end, different cytokines were tested to enhance NK cell commitment (FIG. 2f). Results (Table 1) showed that the use of IL-7 together with SCF and FLT3L during lymphoid commitment brought about a purity of 61% and a yield of 0.21×10^6 CD56⁺CD45⁺ cells from 5.3×10^6 PBC-iPSCs on day 40 of differentiation; using IL-15 increased the purity to 99% and the yield to 0.75×10^6 ; while the combined use of IL-7 and IL-15 gave a purity of 99% and a yield of 7.93×10^6 on day 40, which further increased to 15×10^6 on day 47.

(119) TABLE-US-00001 TABLE 1 Cytokines improve purity and yield of PBC-iPSC derived NK cells

Condition	CD56 ⁺ %	CD45 ⁺ purity	Yield (number)
SCF, SCF, SCF, FLT3L, +IL7	61%	99%	$0.2 \times 10^6 \pm 0.75 \times 10^6 \pm 7.93 \times 10^6 \pm$
Cytokines FLT3L, +IL7 FLT3L, +IL15 IL15	99%	99%	0.18×10^6 0.43×10^6 0.18×10^6 CD45 ⁺)

per 3×10^6 iPSCs per 3×10^6 iPSCs (mean $\pm$ S, (mean $\pm$ S, (mean $\pm$ S, n = 4, day 40) n = 3, day 40) n = 6, day 40)

Generation of $\gamma\delta$ NKT Cells from iPSCs

Cell Culture.

(120) The generated iPSC lines were cultured with mTeSR1 (StemCell Technologies, Vancouver, BC, Canada, <http://www.stemcell.com>) on Matrigel (BD Biosciences, San Diego, CA, <http://www.bdbiosciences.com>)—coated 6-well plates. Cell lines: OP9, (American Type Culture Collection [ATCC], Manassas, VA, <http://www.atcc.org>) were cultured as recommended by ATCC. Cell line OP9-DLL1 (Riken BRC Cell Bank, Ibaraki, Japan, <http://cell.brc.riken.jp/en/>) was cultured in α -MEM (Thermo Fisher Scientific, Waltham, MA, <http://corporate.thermofisher.com>) supplemented with 20% fetal bovine serum (FBS) (HyClone, Logan, UT, <http://www.hyclone.com>).

(121) Generation of $\gamma\delta$ T Cells from GDT-iPSCs.

(122) To generate $\gamma\delta$ T cells from GDT-iPSCs, a previously established two-stage protocol, described above for generating NK cells from iPSCs was used. In the first stage, $1-1.5 \times 10^6$ iPSCs were seeded and differentiated on the overgrown OP9 or OP9-DLL1 cells in α -MEM supplemented with 20% FBS for 12 days. The co-cultures were fed every 4 days by changing half medium. In the second stage, the differentiated cells were harvested from the co-cultures using 1 mg/ml collagenase IV (StemCell Technologies) and TrypLE Express (Thermo Fisher Scientific). OP9 or OP9-DLL1 cells were removed by plastic adherence for 45 minutes and the cell clumps were further removed by 100 μ m cell strainers (BD Biosciences). The remaining non-adherent cells

were then co-cultured with OP9-DLL1 cells grown on T75 flasks using α -MEM containing 20% FBS, 10 ng/ml SCF (PeproTech, Rocky Hill, NJ, <http://www.peprotech.com>), 5 ng/ml FLT3L (PeproTech) together with 5 ng/ml IL-7 (PeproTech) and/or 10 ng/ml IL-15 (PeproTech) for 7 days. Hereafter, the differentiated cells were harvested using Versene (Thermo Fisher Scientific) and cocultured with new OP9-DLL1 cells grown on 6-well plates on a weekly basis for another 3-4 weeks.

(123) To generate $\gamma\delta$ T cells, the previously described above protocol was used for generating NK cells from iPSCs to differentiate the $\gamma\delta$ T cell-derived GDTA/NF-iPSC #1 line described below. On day 33 of differentiation, a homogenous lymphoid population was observed (FIG. 3a), in which 53% was $\gamma\delta$ TCR+CD3+ (FIG. 3b). Most of these $\gamma\delta$ TCR+CD3+ cells were also V γ 9V δ 2+ (FIG. 3c) and CD56+ (FIG. 3d), indicating that these iPSC-derived $\gamma\delta$ T cells were exclusively a V γ 9V δ 2 subset that may have an NK cell-like phenotype. This result tallied well with the findings that 53% of the lymphoid population was CD3+CD56+ and 45% was CD3-CD56+ (FIG. 3e) and that 54% was V δ 2+ (FIG. 3f). On day 47, the lymphoid population became more obvious (FIG. 3g, m). More $\gamma\delta$ TCR+CD3+ cells appeared (FIG. 3h), which remained V γ 9V2+ (FIG. 3i) and CD56+ (FIG. 3j). This change was further confirmed by the increase of CD3+CD56+ population and V δ 2+ population and the decrease of CD3-CD56+ population (FIG. 3k, l). Likewise, when using the non-T cell-derived GDTA/NF-iPSC #3 for differentiation, an obvious lymphoid population was still observed on day 47 (FIG. 3n); however, this population was CD3-CD56+ (FIG. 3p) and there was no $\gamma\delta$ TCR+CD3+ cells (FIG. 3o) or V δ 2+ cells (FIG. 3q) observed. These findings strongly suggest that generation of CD56+ V γ 9V δ 2 T cells from a $\gamma\delta$ T cell-derived iPSC line occurred.

(124) Characterization of Mimetic Innate Immune Cells

(125) Mimetic NK Cells

(126) Flow Cytometry.

(127) To study the change of phenotype during hPSC differentiation, the cells were harvested and stained using antibodies against CD34, CD43, CD45, CD56, TCR $\alpha\beta$, CD3, CD4, CD8, NKp44, NKp46, NKG2D, CD16, NKG2A, CD94, CD158e1/e2, Cd158i, CD158a,h (BD Biosciences and Beckman Coulter, <https://www.beckmancoulter.com>) and analyzed with a FACSCalibur flow cytometer (BD Biosciences).

(128) ELISPOT Assay.

(129) To detect IFN- γ secretion, a Human IFN- γ ELISpotPro kit (MABTECH, Nacka Strand, Sweden, <https://www.mabtech.com>) was used. In brief, 0 to 10×10^4 PBC-iPSC-derived NK cells and 5×10^4 K562 or Raji cells were coincubated on a IFN- γ ELISPOT plate overnight. IFN- γ spots were stained according to the manufacturer's manual. To measure GrB secretion, a Human Granzyme B ELISpot Kit (R&D Systems, Minneapolis, MN, <https://www.mdsystems.com>) was used. In brief, 0 to 1.2×10^4 PBC-iPSC-derived NK cells were incubated with or without 5×10^4 K562 cells on a human GrB microplate for 4 hours. GrB spots were then stained as described in the manufacturer's manual. IFN- γ and GrB spots were counted using an ImmunoSpot Analyzer (CTL, Shaker Heights, OH, <http://www.immunospot.com>).

(130) NK Cells Generated Using IL7 and IL15 Cytokine Combination.

(131) Using the optimized cytokine combination, high-purity lymphoid cells were derived from PBC-iPSC #9 (FIG. 4). These cells showed typical NK cell morphology (FIG. 4a). They were a homogenous population as indicated by the cell images (FIG. 4a) and the scatter plot (FIG. 4b). Most of them are CD56+CD45+CD3-, a typical NK cell phenotype (FIG. 4b). These cells also expressed activating receptors like NKp30, NKp44, NKp46 and NKG2D and inhibitory receptor like CD94:NKG2A (FIG. 4b). They were also CD16+ (FIG. 4b), which is important for antibody-dependent cell-mediated cytotoxicity (ADCC) of NK cells. In term of killer cell immunoglobulin-like receptors (KIR), they showed no or low expression for KIR2DL1, KIR2DL4, KIR2DL5, KIR3DL1, KIR2DS1, KIR2DS4 and KIR3DS1. Only a small fraction of the cells were stained by CD158b (18%) or CD158e/k (13%) antibodies, suggesting that most of the NK cells were negative

for KIR2DL2, KIR2DL3, KIR3DL1 and KIR3DL2 (FIG. 4b). This inhibitory KIR-negative phenotype may potentially benefit the cytotoxicity of these NK cells. Thus, a large number of high-purity NK cells from PBC-iPSCs can be consistently generated using the optimized two-stage protocol.

(132) Mimetic $\gamma\delta$ NK T Cells

(133) Flow Cytometry.

(134) To study phenotypic change during iPSC differentiation, cells were harvested and stained using antibodies against CD3, CD56, $\gamma\delta$ TCR, V δ 2 TCR, V γ 9 TCR, NKp30, NKp44, NKp46, NKG2D, NKG2A, CD94, CD16, FasL, TRAIL, DNAM-1, CD158a,h (KIR2DL1/S1), CD158b (KIR2DL2/L3/S2), CD158f (KIR2DL5), CD158i (KIR2DS4), CD158e1/e2 (KIR3DL1/S1) and CD158e/k (KIR3DL1/L2) (BD Biosciences; Beckman Coulter, <https://www.beckmancoulter.com>; Miltenyi Biotec, <http://www.miltenyibiotec.com>) and analyzed with a FACSCalibur flow cytometer (BD Biosciences).

(135) Expansion of Donor-Derived PB-NK Cells.

(136) To derive PB-NK cells, 2×10^6 PBMCs from healthy donors were co-cultured with 4×10^6 γ -irradiated (100 Gy) K562-mbIL15-41BBL cells in CellGro SCGM serum-free medium supplemented with 10% FBS and 50 IU/mL IL-2 using T75 flasks in upright position. Half medium was replaced with fresh medium and fresh IL-2 was replenished every 2-3 days until day 7. Hereafter, 2×10^6 cells were re-stimulated weekly with 2×10^6 K562-mbIL15-41BBL cells for another two weeks. The cells were harvested on day 21 for experiments.

(137) CD56 is a typical surface marker of NK cells. To find out whether such generated iPSC-derived mimetic $\gamma\delta$ T cells (iPSC- $\gamma\delta$ T cells) also express other surface molecules of NK cells, detailed phenotyping was performed (FIG. 5a). Flow cytometric analysis showed that most iPSC- $\gamma\delta$ T cells highly expressed activating receptors: NKG2D and DNAM-1; natural cytotoxicity receptors (NCRs): NKp30, NKp44 and NKp46; and death-inducing ligand: TRAIL (FIG. 5a). In addition, 41% expressed CD16, a surface molecule mediates antibody-dependent cell-mediated cytotoxicity (ADCC); 5% expressed FasL, another death-inducing ligand (FIG. 5a). This surface molecule expression profile of iPSC- $\gamma\delta$ T cells closely resembled that of expanded peripheral blood NK (PB-NK) cells (FIG. 5c), which had low-level TRAIL expression. In contrast, although the expanded peripheral blood $\gamma\delta$ T (PB- $\gamma\delta$ T) cells also expressed NKG2D and DNAM-1, they expressed no NCRs, death-inducing ligands and CD16 (FIG. 5b). In terms of inhibitory receptors and killer cell immunoglobulin-like receptors (KIRs), iPSC- $\gamma\delta$ T cells expressed CD94/NKG2A receptor like PB-NK cells and PB- $\gamma\delta$ T cells; however, unlike PB-NK cells and PB- $\gamma\delta$ T cells, iPSC- $\gamma\delta$ T cells did not express KIRs (FIG. 6a-c), which makes them unrestricted by recipient's HLA phenotype. Thus, the iPSC- $\gamma\delta$ T cells are armed with both sets of “cancer-fighting” molecules observed in PB- $\gamma\delta$ T cells and PB-NK cells and such novel GDT-iPSC-derived “ $\gamma\delta$ NKT cells” may fight against a broad spectrum of cancer cells.

(138) Cancer Immunotherapy Using Mimetic Innate Immune Cells

(139) Treatment with Mimetic NK Cells

(140) Cell Culture.

(141) K562 cells (ATCC), K562-mb15-41BBL cells (kindly provided by Prof. Dario Campana, Department of Paediatrics, National University of Singapore, Singapore) and Raji cells (ATCC) were cultured with RPMI 1640 (Thermo Fisher Scientific) supplemented with 10% FBS.

(142) Cytotoxicity and ADCC Assay.

(143) To detect direct cytotoxicity of PBC-iPSC-derived NK cells against target cells, a flow cytometry method was used. In brief, 0 to 0.5×10^6 NK cells were co-incubated with 2×10^4 carboxyfluorescein diacetate succinimidyl ester (CFSE; Thermo Fisher Scientific)—labelled K562 or Raji cells for 4-6 hours. The samples were then stained on ice with 7-Amino-Actinomycin D (7-AAD, BD Biosciences) for 10 minutes. After washing, the target cell death was assessed by FACSCalibur flow cytometer based on the percentage of 7-AAD-stained cells in the

CFSE-positive population. To evaluate the ADCC function of PBC-iPSC-derived NK cells, co-cultures of NK cells and CFSE-labelled Raji cells were set up in the presence of human IgG1 (Sigma-Aldrich), anti-CD20-mIgG1 (InvivoGen, San Diego, CA, <http://www.invivogen.com>) or anti-CD20-hIgG1 (InvivoGen) of various concentrations, the target cell death was measured by flow cytometer as described above.

(144) Functions of PBC-iPSC-Derived NK Cells.

(145) Secreting cytokines such as interferon- γ (IFN- γ) upon activation is an important functional feature of NK cells. To investigate IFN- γ secretion, PBC-iPSC-derived NK cells were co-cultured with stimulatory cells and an Enzyme-Linked ImmunoSpot (ELISPOT) assay was then used to detect IFN- γ -secreting NK cells. As shown in FIGS. 7a and 7b, K562, an NK cell-sensitive cell line efficiently stimulated IFN- γ secretion by NK cells derived from PBC-iPSC #9; in contrast, Raji, an NK cell-resistant cell line showed much lesser efficiency, suggesting that these PBC-iPSC-derived NK cells are capable of secreting cytokine in response to stimulation.

(146) Cytotoxicity is another hallmark feature of NK cells, which depends on the secretion of cytotoxic molecules like granzyme B (GrB). ELISPOT result showed that a high frequency of PBC-iPSC-derived NK cells secreted GrB upon stimulation by K562 (FIG. 7c, d), which further confirms the functional competence of these iPSC-derived NK cells. More importantly, these iPSC-derived NK cells showed a similar direct killing profile as primary NK cells. As demonstrated in FIGS. 7e and 7f, they were able to directly kill the NK-sensitive leukemia cells K562, but not the NK-insensitive lymphoma cells Raji. Interestingly, by exploiting the ADCC function of these iPSC-derived NK cells, Raji cells were still able to be killed (FIG. 7g, h). As shown in FIGS. 7g and 7h, NK cells alone were not able to kill Raji cells; neither with the addition of human IgG1 or anti-CD20-mIgG1; however, in the presence of anti-CD20-hIgG1, obvious cytotoxicity on Raji cells were observed. Thus, the ADCC function of these PBC-iPSC-derived NK cells can significantly broaden their killing spectrum of malignant cells.

Treatment with Mimetic $\gamma\delta$ NK T Cells

(147) Cell Culture.

(148) Cell lines: SK-OV-3, SW480, MCF-7 and BT474 (American Type Culture Collection [ATCC], Manassas, VA, <http://www.atcc.org>) were cultured as recommended by ATCC.

(149) Cytotoxicity Assay.

(150) To detect cytotoxicity of iPSC-derived $\gamma\delta$ NK T cells against target cells, a flow cytometry-based method was used. In brief, 0 to 0.5×10^6 $\gamma\delta$ NK T cells were co-cultured with 2×10^4 carboxyfluorescein diacetate succinimidyl ester (CFSE; Thermo Fisher Scientific)—labelled cancer cells at various effector to target (E:T) ratios for 4-6 hours. Samples were then stained on ice with 7-Amino-Actinomycin D (7-AAD, BD Biosciences) for 10 minutes. After washing, target cell death was assessed with flow cytometer by the percentage of 7-AAD-stained cells in CFSE-positive population.

(151) To test the cytotoxicity of $\gamma\delta$ NKT cells, four different solid tumor cell lines were used as target cells: MCF7 (breast adenocarcinoma), SW480 (colorectal adenocarcinoma), BT474 (breast ductal carcinoma) and SK-OV-3 (ovary adenocarcinoma). Results showed that the $\gamma\delta$ NKT cells were able to kill these cancer cell lines (FIG. 8). In some cases, the cytotoxicity was observed even at very low E:T ratios (FIG. 8), suggesting the high efficacy of the $\gamma\delta$ NKT cells.

(152) Generation of iPSCs from Peripheral Blood Cells

(153) Generation of iPSCs from PBCs.

(154) To generate iPSCs from PBCs, frozen peripheral blood mononuclear cells (PBMCs) from healthy donors (StemCell Technologies) were thawed and cultured with 5 μ g/ml phytohemagglutinin (PHA; Sigma-Aldrich, St Louis, MO, <http://www.sigmaaldrich.com>) for 2 days in complete RPMI 1640 medium, which is composed of RPMI 1640 (Thermo Fisher Scientific), 10% heat-inactivated human serum AB (Gemini Bio-Products, West Sacramento, CA, <http://www.gembio.com>), 2 mM L-glutamine (Thermo Fisher Scientific), 0.1 mM nonessential

amino acids (Thermo Fisher Scientific) and 0.1 mM 2-mercaptoethanol (Thermo Fisher Scientific). The cultured blood cells were then transduced with Sendai reprogramming vectors from a CytoTune iPS 2.0 Sendai Reprogramming Kit (Thermo Fisher Scientific) at MOI of 5:5:3 (KOS, hc-Myc, hKlf4) in complete RPMI 1640 medium containing 10 ng/ml of IL-2, IL-7 and IL-15 overnight. The transduced cells were then washed and cultured for 5 days before seeding to a six-well plate grown with mitomycin C (Sigma-Aldrich)—treated mouse embryonic fibroblasts (mEFs). Half of medium was replaced on day 1 to 3 after seeding with iPSC medium, which is composed of DMEM/F12 (Thermo Fisher Scientific), 20% knockout serum replacement (Thermo Fisher Scientific), 2 mM L-glutamine, 1% nonessential amino acids, 0.1 mM 2-mercaptoethanol and 5 ng/ml basic fibroblast growth factor (PeproTech). Three to four weeks after seeding, the resulting iPSCs were first expanded on mEFs in iPSC medium and later on Matrigel-coated plates in mTeSR1.

(155) Immunostainina.

(156) To detect pluripotency markers, PBC-iPSCs were fixed in 4% paraformaldehyde (Sigma-Aldrich) at room temperature for 15 min followed by blocking with 5% bovine serum albumin for 1 hour. The cells were then incubated with primary antibodies against SSEA-4, TRA-1-60, TRA-1-81 (Thermo Fisher Scientific) for 1 hour at room temperature. After washing, the cells were incubated with Alexa Fluor 594-conjugated goat anti-mouse IgG (H+L) antibody (Thermo Fisher Scientific) at room temperature for 1 hour for visualization under fluorescence microscope. To detect alkaline phosphatase (AP), Alkaline Phosphatase Live Stain (Thermo Fisher Scientific) was used according to the manufacturer's protocol. To demonstrate the feasibility of using PBC-iPSCs for NK cell generation, iPSC lines from PBCs using Sendai viral vectors carrying the reprogramming factors were first generated (FIG. 2a). These resulting PBC-iPSCs showed typical hPSC morphology and phenotype (FIG. 2a).

(157) Expansion of $\gamma\delta$ T Cells.

(158) To expand $\gamma\delta$ T cells for iPSC generation, frozen PBMCs from a healthy donor (StemCell Technologies) were thawed and cultured in PBMC culture medium [OpTmizer CTS T-Cell Expansion SFM (Thermo Fisher Scientific) supplemented with 10% heat-inactivated human AB serum (Gemini Bio-Products, West Sacramento, CA, <http://www.gembio.com>) and 10 ng/ml IL-2 (Thermo Fisher Scientific)] containing 5 μ M Zometa (Novartis). Half medium was replaced with fresh PBMC culture medium without Zometa every 2-3 days.

(159) Generation of GDT-iPSCs.

(160) To generate iPSCs from the expanded $\gamma\delta$ T cells, the cultured PBMCs were transduced on day 7 with Sendai reprogramming vectors from a CytoTune iPS 2.0 Sendai Reprogramming Kit (Thermo Fisher Scientific) at MOI of 5:5:3 (KOS, hc-Myc, hKlf4) overnight. The transduced cells were then washed and cultured for 5 days before seeding to a 6-well plate grown with mitomycin C (Sigma-Aldrich)—treated mEFs. Half medium was replaced on day 1 to 3 after seeding with iPSC medium, which is composed of DMEM/F12 (Thermo Fisher Scientific), 20% knockout serum replacement (Thermo Fisher Scientific), 2 mM L-glutamine, 1% nonessential amino acids, 0.1 mM 2-mercaptoethanol and 5 ng/ml basic fibroblast growth factor (PeproTech). Three to four weeks after seeding, the resulting iPSC colonies were picked up and expanded on Matrigel-coated plates in mTeSR1.

(161) To generate iPSCs from $\gamma\delta$ T cells using a non-viral method, PBMCs were cultured with Zometa and IL-2 as described above and used on day 7 or day 13. On day 0, episomal reprogramming vectors from a Epi5 Episomal iPSC Reprogramming Kit (Thermo Fisher Scientific) were delivered into the cultured PBMCs via nucleofection using a Amaxa Nucleofector 2b (Lonza, <http://www.lonza.com>). The nucleofected cells were then seeded on mitomycin C-inactivated mEFs. On day 2, the cells were adapted to a 1:1 mixture of PBMC culture medium: iPSC medium. From day 3 on, the cells were cultured in iPSC medium, which was changed every other day. Two to four weeks after seeding, iPSC colonies were picked up and expanded in

Matrigel-coated plates in mTeSR1.

(162) TCRB and TCRG Gene Clonality Assays.

(163) Genomic DNA was isolated from iPSCs using a DNeasy Blood and Tissue Kit (Qiagen, <https://www.qiagen.com>) according to the manufacturer's instruction. To detect TCRG and TCR γ gene rearrangement in genomic DNA, PCR was carried out with master mixes provided in TCRB and TCRG Gene Clonality Assay kits (Invivoscribe Technologies, San Diego, CA, <http://www.invivoscribe.com>) and AmpliTaq Gold DNA polymerase (Thermo Fisher Scientific) using the following program: 95° C. for 7 minutes; 35 amplification cycles (95° C. for 45 seconds, 60° C. for 45 seconds, 72° C. for 90 seconds); and final extension of 72° C. for 10 minutes before holding at 15° C. PCR products were separated by electrophoresis in 2% MetaPhor Agarose (Lonza, <http://www.lonza.com>) gel.

(164) To activate and expand $\gamma\delta$ T cells for iPSC generation, peripheral blood mononuclear cells (PBMCs) from a healthy donor was cultured using Zometa and IL-2. Total cell number and cell clumps in the PBMC culture increased over time (FIG. 9a), indicating a successful expansion of $\gamma\delta$ T cells. Flow cytometric analysis further confirmed that up to 61% of the lymphocytes in a 7-day culture were $\gamma\delta$ T cells (FIG. 9b). This $\gamma\delta$ T cell-enriched PBMC culture was then used for iPSC generation.

(165) To derive iPSC lines from $\gamma\delta$ T cells, two reprogramming protocols were used. In one protocol, integration-free Sendai viral vectors carrying the reprogramming factor genes to transduce the cultured PBMCs was initially used. As observed on day 6 after transduction, most transduced cells survived (FIG. 10a). These cells were then seeded on mitomycin C-treated mouse embryonic fibroblasts (mEFs). iPSC colonies (FIG. 10b) started to appear as early as 9 days after seeding. These colonies could be picked up 3 weeks after seeding and directly expanded on Matrigel-coated plates in mTeSR1 medium (FIG. 10c).

(166) In another protocol, nucleofection was used to deliver episomal reprogramming vectors into the cultured PBMCs and established three iPSC lines: GDTA/NF-iPSC #1, GDTA/NF-iPSC #2 and GDTA/NF-iPSC #3 (FIG. 10d-f). To identify iPSC lines derived from $\gamma\delta$ T cells, TCRG gene clonality assay (FIG. 10g) and TCRB gene clonality assay were performed. Two iPSC lines, GDTA/NF-iPSC #1 and GDTA/NF-iPSC #2 were confirmed to be derived from $\gamma\delta$ T cells due to the presence of rearranged TCRG gene, whereas one iPSC line, GDTA/NF-iPSC #3 was from a non-T cell due to the absence of both rearranged TCRB and TCRG genes. Like the non-T cell-derived GDTA/NF-iPSC #3, the $\gamma\delta$ T cell-derived GDTA/NF-iPSC #1 and GDTA/NF-iPSC #2 expressed no $\gamma\delta$ TCRs and CD3 (FIG. 9h), suggesting the thorough reprogramming of $\gamma\delta$ T cells.

(167) For the first time the derivation of iPSCs from $\gamma\delta$ T cells and the generation of $\gamma\delta$ T cells using such $\gamma\delta$ T cell-derived iPSCs (GDT-iPSCs) has been demonstrated. Moreover, by combining these novel GDT-iPSC lines with the above described differentiation protocol for generating NK cells from iPSCs, $\gamma\delta$ T cells that highly expressed the “cancer-fighting” molecules of NK cells were generated and designated this unique cell type as “ $\gamma\delta$ natural killer T (NKT) cells”.

(168) Expansion of Mimetic Innate Immune Cells

(169) Expansion of Fresh and Cryopreserved PBC-iPSC-Derived NK Cells.

(170) Generating sufficient functional cells is a prerequisite to translate these iPSC-derived NK cells into clinical use. One possible way to obtain therapy-scale NK cells from iPSCs is to scale up the differentiation cultures, which, however, is not cost-effective. A more practical way is to expand the iPSC-derived NK cells using feeder cells. To investigate this possibility, various numbers of NK cells generated from PBC-iPSC #9 were co-cultured with irradiated K562-mb15-41BBL cells at an NK cell: feeder ratio of 1:10 in gas-permeable G-Rex10 flasks. The results showed that these iPSC-derived NK cells were quickly expanded during a 14-day co-culture (FIGS. 11a, b). With a starting NK cell number of 10, a 74-fold expansion was observed on day 9, although no further expansion was achieved by extending the co-culture to 14 days (FIG. 11a). Interestingly, although there was no obvious change in phenotype except the down-regulation of CD16

expression (FIG. 11c), these expanded NK cells became more potent in killing K562 cells (FIG. 11d), probably due to their further functional maturation during expansion. These data indicate that it is feasible to produce functional iPSC-derived NK cells in clinical-scale through expansion with feeder cells.

(171) Shipping cellular therapeutics from a centralized manufacturing site to a clinical site for injection without compromising product quality is crucial for the clinical efficacy of live cell products. Transporting conventional NK cell products in cryopreserved form can significantly reduce the viability and potency of NK cells. This was also true for the PBC-iPSC-derived NK cells (FIGS. 11e, g). The freeze/thaw procedure significantly affected the viability of expanded PBC-iPSC-derived NK cells (FIG. 11e), although the potency of these expanded NK cells was preserved to a certain extent after freeze/thaw procedure (FIG. 11g). To overcome these viability and potency issues caused by cryopreservation, a possible solution: the PBC-iPSC-derived NK cells are first cryopreserved before expansion at a centralized manufacturing site and shipped to a clinical site in cryopreserved form was proposed; these cryopreserved NK cells are then thawed, expanded and injected at the clinical site. The feasibility of such solution depends on the ability to expand the cryopreserved NK cells. As shown in FIG. 11f, the cryopreserved NK cells remained very expandable; starting from 10×10^6 cryopreserved NK cells, a 38.5-fold expansion could still be achieved on day 9. Most importantly, these expanded cryopreserved NK cells showed comparable cytotoxic potency as those expanded from non-cryopreserved fresh NK cells (FIG. 11g). These results suggest that transporting pre-expansion NK cells is a practical solution for the logistics of PBC-iPSC-derived NK cell therapeutics.

(172) The yield was up to 15×10^6 NK cells per 3×10^6 PBC-iPSCs with 99% purity. These PBC-iPSC-derived NK cells could be expanded by 74-fold in 9 days and by 38.5-fold after cryopreservation using a feeder cell line. These expanded NK cells became more potent functionally as shown by the increased cytotoxicity against K562 cells. Thus, the feasibility of generating NK cell therapeutics from PBC-iPSCs using an industry-compatible production scheme is demonstrated.

(173) It should be further appreciated by the person skilled in the art that variations and combinations of features described above, not being alternatives or substitutes, may be combined to form yet further embodiments falling within the intended scope of the invention.

Claims

1. A method of generating gamma-delta natural killer T cells ($\gamma\delta$ NKT cells), the method comprising: (a) expanding $\gamma\delta$ T cells of human peripheral blood cells in PBMC culture medium containing an amino-bisphosphonate and interleukin 2 (IL2); (b) transducing the expanded $\gamma\delta$ T cells with reprogramming transcription factors to generate induced pluripotent stem cells (iPSCs); (c) screening the TCRG and TCRD gene configuration of the iPSCs to identify $\gamma\delta$ T cell-derived iPSCs; (d) co-culturing $\gamma\delta$ T cell-derived iPSCs with a stromal cell line deficient in expressing macrophage colony stimulating factor (M-CSF) to generate innate immune cell progenitors; (e) co-culturing the innate immune cell progenitors with a stromal cell line deficient in expressing M-CSF and ectopically expressing Notch ligand, Delta like 1 (DLL1) in a media comprising stem cell factor (SCF), Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 7 (IL7) and interleukin 15 (IL15) to generate differentiated gamma-delta NKT cells; (f) passaging the differentiated gamma-delta NKT cells weekly for 3 to 5 weeks to obtain gamma-delta NKT cells expressing gamma-delta TCRs ($\gamma\delta$ TCRs); and (g) detecting the expression profile of the gamma-delta NKT cells obtained in step (f); wherein the $\gamma\delta$ NKT cells expressing gamma-delta TCRs ($\gamma\delta$ TCRs), express NKp30 in 99% of the cells, NKp44 in 99% of the cells, NKp46 in 97% of the cells, NKG2D in 84% of the cells, DNAM-1 in 97% of the cells, TRAIL in 89% of the cells, and NKG2A/CD94 in 83% of the cells in combination with expression of at least any one of CD56, CD16, FASL and less than 5%

expression of killer cell immunoglobulin-like receptors (KIR) are $\gamma\delta$ NKT cells.

2. The method according to claim 1, wherein the $\gamma\delta$ T cell-derived iPSCs in (c) are co-cultured with the stromal cell line deficient in expressing M-CSF for 12 days.

3. The method according to claim 1, wherein the media comprises Fetal Bovine Serum (FBS); alpha Minimum Essential Medium (aMEM); or a combination thereof.

4. The method according to claim 1, wherein the amino-bisphosphonate is zoledronic acid or salts thereof.

5. The method according to claim 1, wherein the iPSCs generated in step (b) comprises a TCR gene arranged in a TCRG gene configuration and a TCR gene arranged in a TCRD gene configuration.

6. The method according to claim 1, wherein the method further comprises expanding gamma-delta NKT cells in the presence of feeder cells in a media comprising stem cell factor (SCF); Fetal Bovine Serum (FBS); Fms-related tyrosine kinase 3 ligand (FLT3L), interleukin 15 (IL15) and interleukin 7 (IL7).
